7. Risk Framework — 14 Risks with Mitigation Strategies

> **Risk Philosophy:** Every risk has a mitigation strategy, a trigger condition, and a rollback plan. We don't just identify risks — we pre-build the response. The brief specifically asks for "risk mitigation plan (human-in-the-loop, rollback strategies)" — this section delivers both.

7.1 Risk Heat Map

#	Risk	Likelihood	Impact	Severity	Phase	Mitigation Summary
R1	**API Access Delays**	HIGH	HIGH	CRITICAL	Phase 1	Start provisioning Day 1. Demo mode unblocks development. Parallel work on agents that don't need live APIs.
R2	**Agent Hallucination / Wrong Actions**	MEDIUM	HIGH	CRITICAL	All	HITL gates on all writes. Guardrails (list validation, max actions/run). Tracing for audit. Graduation path.
R3	**Low Adoption by Team**	MEDIUM	HIGH	HIGH	All	Survey shows 89% already use AI tools. Role-specific training. Champions program. Quick wins build trust first.
R4	**ClickUp API Rate Limits**	MEDIUM	MEDIUM	HIGH	Phase 1	Request queue with exponential backoff. Batch operations where possible.

						Cache frequently-read data (lists, spaces).
R5	**Frame.io No MCP Available**	HIGH	MEDIUM	HIGH	Phase 1	Custom Function Tools (4 API wrappers) already designed in Batch 4. No dependency on MCP. Webhook handlers for real-time events.
R6	**PPM Burnout During Transition**	HIGH	MEDIUM	HIGH	Phase 1	PPMs at 121% utilization. Prioritize ED-1 and ED-5 (their highest-impact agents). Temporary workload redistribution during pilot.
R7	**OpenAI API Cost Overrun**	LOW	MEDIUM	MEDIUM	Phase 2+	Budget alerts at 80% threshold. Model selection: GPT-4o-mini for routine, GPT-4o for complex. Token usage monitoring per agent.
R8	**Scope Creep**	MEDIUM	MEDIUM	MEDIUM	All	16 wish-list tools are the scope. Phase exit criteria must be met before new requests. Change request process via Slack.

R9	**Data Privacy / Client Content Exposure**	LOW	HIGH	MEDIUM	All	No client video content sent to LLMs. Only metadata (titles, descriptions, comments). OpenAI data processing agreement. Notion scoped to internal DBs only.
R10	**MCP Server Instability**	LOW	MEDIUM	MEDIUM	Phase 1	Fallback to direct API Function Tools if MCP server fails. Health checks every 60s. Auto-restart via Docker.
R11	**Key Person Dependency (AI Gen X)**	LOW	MEDIUM	LOW	All	All code in shared repo. Documentation per agent. Runbooks for operations. Knowledge transfer in Phase 3.
R12	**Tool Stack Changes (DFM switches tools)**	LOW	LOW	LOW	Phase 3	Connector abstraction layer. Swapping ClickUp for another PM tool = new connector, same agent logic.
R13	**OpenAI SDK Breaking Changes**	LOW	LOW	LOW	Phase 2+	Pin SDK version. Test upgrades in staging. SDK is production-grade with stable API.
R14	**Editor Resistance (54 people)**	MEDIUM	LOW	LOW	Phase 1	Editors are "power

		users" (survey: 89% use AI). Agents help them (faster assignments, less idle time), not replace them. Frame as "less waiting, more editing."

7.2 Critical Risk Deep Dives

R1: API Access Delays — Detailed Mitigation

Scenario	Response	Impact
ClickUp API key delayed >2 weeks	Continue development in demo mode. All agents work against mock data. No schedule impact on agent code — only live testing delayed.	Low
Slack Bot Token delayed	Use Slack Incoming Webhooks (no bot token needed) for Phase 0 templates. Bot token only needed for Phase 1 agent gateway.	Low
Frame.io API access denied	ED-1 QC Checker works on exported metadata (CSV/JSON). Webhook integration deferred. Manual upload bridge.	Medium
All API access delayed >30 days	**Escalation:** AI Gen X escalates to DFM leadership. Phase 0 quick wins continue unblocked. Phase 1 start date shifts but Phase 0 value is preserved.	High

R2: Agent Hallucination — Defense-in-Depth

Layer	Defense	Implementation
1	**Agent Instructions**	Precise, role-specific instructions (from Batch 3 playbooks). "You are a Producer agent. You ONLY create tasks in the following lists: [list_ids]. You NEVER delete tasks."
2	**Guardrails**	OpenAI SDK guardrails: input validation (reject off-topic), output validation (schema check). Max 3 write operations per run.
3	**Tool Constraints**	Tools validate parameters before execution. `create_task` checks: list_id in allowed list, assignee in team roster, due date in valid range.
4	**HITL Gate**	All write operations require human approval (Slack button: ✅ Approve / ❌ Reject / ✏️ Edit). Shows exactly what will be created/modified.
5	**Tracing & Audit**	Every agent run logged: input, reasoning steps, tool calls, outputs, HITL decisions. Searchable in dashboard. 90-day retention.

6	**Rollback**	Every write operation stores the "before" state. One-click rollback via dashboard or Slack command: `/rollback [trace_id]`.

R3: Low Adoption — Survey-Calibrated Response

> **Survey Data (73 respondents):** 89% of DFM team already uses AI tools. Editors are power users. CDs have highest request rate for new tools. PPMs are most time-constrained. This is NOT a team that resists AI — it's a team that wants BETTER AI tools.

Role	AI Adoption	Risk Level	Adoption Strategy
Editors	89% use AI	LOW	Frame as "less idle time, faster assignments." Agents serve them, not replace them.
CDs	100% ChatGPT	LOW	Highest request rate. CD-3 /roast command is fun and viral — drives organic adoption.
Producers	50% ChatGPT	MEDIUM	PR-4 saves the most time (ClickUp 38%→18%). Demo the time savings first. Champions program.
PCs	67% ChatGPT	MEDIUM	ED-2 capacity assignment is their biggest pain point. Show it working on real data.
PPMs	Varies	MEDIUM	At 121% utilization, they NEED help. ED-1 QC Checker is positioned as "burnout relief."

8. Human-in-the-Loop Governance Model

HITL GOVERNANCE · 7 Gates · 3 Trust Levels · Graduation Path

8.1 Three Trust Levels

Level	Behavior	When Applied	User Experience
ALWAYS_CONFIRM	Agent proposes action → waits for human approval before executing	All agents at initial deployment. All write operations.	Slack message: "PR-4 wants to create task: [details]. ✅ ❌ ✏️"
NOTIFY_ONLY	Agent executes action → notifies human after	After graduation (≥95% approval rate over 50+ runs)	Slack message: "PR-4 created task: [details]. ↩️ Undo"
AUTONOMOUS	Agent executes silently. Logged but no notification.	Only for read operations and low-risk writes (e.g., adding comments). Never for task creation or status changes without explicit graduation.	No notification. Visible in dashboard.

8.2 Seven HITL Gates

#	Gate	Agent(s)	What's Gated	Initial Level
G1	**Task Creation**	PR-4	Creating new ClickUp tasks from natural language	ALWAYS_CONFIRM
G2	**Editor Assignment**	ED-2	Assigning editors to tasks based on capacity algorithm	ALWAYS_CONFIRM

G3	**Status Changes**	PR-4, ED-2	Moving tasks between statuses (e.g., "In Progress" → "Review")	ALWAYS_CONFIRM
G4	**Client Communication**	PR-3, PR-2	Sending messages to client-facing Slack channels	ALWAYS_CONFIRM
G5	**QC Verdicts**	ED-1	Marking videos as pass/fail on technical quality checks	ALWAYS_CONFIRM
G6	**Content Publishing**	YouTube agent	Publishing videos or updating metadata on YouTube	ALWAYS_CONFIRM
G7	**Financial Operations**	Finance agent	Creating invoices, updating billing records in QuickBooks	ALWAYS_CONFIRM

8.3 Graduation Criteria

Metric	ALWAYS_CONFIRM → NOTIFY_ONLY	NOTIFY_ONLY → AUTONOMOUS
Approval Rate	≥95% over 50+ runs	≥99% over 200+ runs
Error Rate	<2% (actions requiring rollback)	<0.5%
Time in Level	Minimum 2 weeks	Minimum 4 weeks
User Satisfaction	Role lead confirms comfort	Leadership sign-off required
Rollback Frequency	<5% of actions rolled back	<1% of actions rolled back

> **Important:** Gates G4 (Client Communication), G6 (Content Publishing), and G7 (Financial Operations) may **never** graduate to AUTONOMOUS. These remain at NOTIFY_ONLY permanently unless DFM leadership explicitly approves. Client-facing and financial actions always have a human in the loop.

8.4 Rollback Mechanism

Every write operation stores a rollback record:

Field	Description
`trace_id`	Unique identifier for the agent run
`action_type`	create_task, update_status, send_message, etc.
`before_state`	JSON snapshot of the entity before modification
`after_state`	JSON snapshot after modification

`rollback_action`	API call to reverse the change (e.g., delete task, revert status)
`rolled_back`	Boolean — whether rollback was executed
`rolled_back_by`	User who triggered rollback

Rollback triggers:

- Slack command: `/rollback [trace_id]`
- Dashboard: "Undo" button on any recent action
- Automatic: If downstream validation fails (e.g., task created in wrong list)

9. Training & Adoption Framework

> **Brief requirement:** "Training and adoption plan for each role." This section provides role-specific, phase-aligned training with survey-calibrated content.

9.1 Training Philosophy

#	Principle	Implementation
1	**Role-Specific**	Producers learn PR-4 and PR-3. CDs learn CD-1 through CD-5. No one-size-fits-all sessions.
2	**Just-in-Time**	Training happens when the agent deploys, not weeks before. Reduces cognitive load.
3	**Show, Don't Tell**	Every training session includes a live demo with real (or realistic mock) data from DFM's pipeline.
4	**Champions First**	Train 1 champion per role on pilot pod. Champions train their peers during rollout.
5	**Async Materials**	Loom recordings of every training session. Notion runbooks per agent. Slack pinned quick-reference guides.

9.2 Role-Specific Training Schedule

Role	Phase	Agents	Duration	Format	Content
Producers	Phase 0	PR-3 (templates)	15 min	Slack + Loom	Slash commands, template selection, when to use which template
Producers	Phase 1	PR-4	30 min	Live + Loom	Natural language → task creation. HITL approval flow. Edge cases. "What to say vs. what not to say."
Producers	Phase 2	PR-1, PR-2	20 min	Live + Loom	Timeline predictions, sentiment dashboard, interpreting scores
PCs	Phase 1	ED-2	30 min	Live + Loom	Capacity algorithm explanation. Override process. Assignment review.
PPMs	Phase 1	ED-1, ED-5	30 min	Live + Loom	QC checklist interpretation. Sentiment scores. When to override. Escalation paths.
CDs	Phase 0	CD-4 (Claude)	30 min	Live + Loom	

					Claude Projects setup. Prompt engineering for ideation. Knowledge base updates.
CDs	Phase 2	CD-1 to CD-5	45 min	Live + Loom	Retention scoring interpretation. Competitor data. /roast command. Voice matching. Outlier detection.
Editors	Phase 1	ED-1 (consumer)	15 min	Slack + Loom	How QC feedback appears. How to respond. What's automated vs. what's not.
All Roles	Phase 3	All agents	60 min	Live workshop	Full system overview. Cross-agent flows. Dashboard walkthrough. Q&A.

9.3 Champions Program

Role	Recommended Champion	Why	Responsibilities
Producer	Pilot pod Producer	First to use PR-4. Builds institutional knowledge.	Train Pod 2/3 Producers. Report bugs. Suggest improvements.
PC	Pilot pod PC (Garrett or Mariana)	Closest to editor assignment workflow.	Validate ED-2 assignments. Train other PCs.
PPM	Mahmoud or Martina	Named in pipeline as PPM reviewers. Highest utilization = most motivated.	Validate QC results. Calibrate sentiment thresholds.
CD	Hannah or Matias	Named as creative reviewers. 100% AI adoption (survey).	Test CD tools. Provide creative quality feedback.
Editor	Senior editor (TBD)	Editors are power users. Need a voice in the process.	Represent editor perspective. Test ED-3/ ED-4 outputs.

9.4 Adoption Metrics

Metric	30-Day Target	90-Day Target	EOY Target	Measurement
Daily Active Agent Users	5 (pilot pod)	15 (3 pods)	20+ (all pods)	Agent trace logs (unique users/day)
Agent Runs per Week	50	200	500+	Observability dashboard

HITL Approval Rate	≥85%	≥95%	≥98%	HITL logs
Training Completion	100% pilot pod	100% all pods	100% + refresher	Attendance + Loom view count
User Satisfaction (NPS)	≥7/10	≥8/10	≥9/10	Monthly survey (5 questions)

10. Engagement Model & Working Cadence

> **Brief requirement (10% Engagement Fit):** "How we'd work together — cadence, communication, decision-making." DFM operates async-first, Slack-heavy. This engagement model matches that exactly.

10.1 Communication Channels

Channel	Purpose	Cadence	Participants
#ai-automation (Slack)	Primary async channel	Daily — all updates, questions, decisions	AI Gen X + DFM stakeholder + pod leads
#ai-automation-alerts (Slack)	Agent alerts & HITL requests	Real-time — automated	Relevant role leads
Biweekly Steering Call	Progress review, blockers, priorities	Every 2 weeks, 30 min	AI Gen X lead + Brendan/Yazan/Emily
Weekly Written Update	Status report	Every Friday, posted to #ai-automation	AI Gen X → DFM
GitHub Repo	All code, PRs, issues	Continuous	AI Gen X engineers + DFM tech lead (if any)
Observability Dashboard	Agent performance, traces, KPIs	Always accessible	All DFM stakeholders

10.2 Decision-Making Framework

Decision Type	Who Decides	Timeline	Process
Agent instruction changes	AI Gen X	Same day	Implement, notify in Slack, log in PR
New tool/connector addition	AI Gen X + DFM stakeholder	48 hours	Propose in Slack, DFM approves, implement
HITL graduation (trust level change)	DFM stakeholder	1 week	AI Gen X presents data, DFM decides
Phase transition (e.g., Phase 1 → 2)	DFM leadership	Steering call	

			Exit criteria review, leadership sign-off
Scope change / new wish-list item	DFM leadership	Steering call	Change request form, impact assessment, prioritization
Emergency rollback	Anyone	Immediate	`/rollback [trace_id]` in Slack. No approval needed.

10.3 AI Gen X Team Composition

Role	Count	Allocation	Responsibilities
Project Lead	1	50%	DFM stakeholder interface. Steering calls. Architecture decisions. Quality assurance.
Agent Engineer	1	100%	Agent development (OpenAI SDK). MCP servers. Connectors. HITL implementation.
Integration Engineer	1	75%	ClickUp, Slack, Notion, Frame.io API integrations. Make scenario development. Webhook handlers.
ML/AI Specialist	1	25% (Phase 2+)	Retention scoring model. Sentiment analysis. Voice matching. Competitor analysis pipelines.

10.4 Weekly Written Update Template

📋 **AI Automation Weekly Update — Week [N]**

✅ **Completed This Week:**
• [Deliverable 1] — [status]
• [Deliverable 2] — [status]

🔄 **In Progress:**
• [Task 1] — [% complete, ETA]
• [Task 2] — [% complete, ETA]

🚧 **Blockers:**
• [Blocker 1] — [owner, action needed]

📊 **Metrics:**
• Agent runs this week: [N]
• HITL approval rate: [N%]
• Errors: [N]

📅 **Next Week:**

- [Planned deliverable 1]
- [Planned deliverable 2]

11. Success Metrics & KPI Dashboard

11.1 Primary KPIs — Tied to Dream Allocations

#	KPI	Baseline	30-Day	90-Day	EOY	Source
1	**Clients per Pod**	~4	4	5	**7**	ClickUp client count
2	**Producer Coordination Overhead**	74.5%	70%	55%	**41%**	Time audit
3	**PC Coordination Overhead**	~70%	65%	50%	**40%**	Time audit
4	**PPM Utilization**	121.2%	115%	105%	**100%**	Time audit
5	**CD Preproduction Time**	29%	32%	40%	**50%**	Time audit
6	**Editor Core Editing Time**	66%	68%	73%	**79%**	Time audit
7	**Annual Capacity Value Recovered**	$0	$15K run rate	$200K run rate	**$883K**	Batch 5 model
8	**Agent Adoption Rate**	0%	100% pilot	100% all pods	**100%**	Agent trace logs

11.2 Agent-Specific KPIs

Agent	KPI	Target	Measurement
PR-4	Tasks created via agent	≥80% of new tasks	ClickUp audit: agent-created vs. manual
PR-3	Template usage rate	≥60% of Slack messages use templates	Slack analytics
ED-2	Assignment accuracy	≥90% match to optimal	PC override rate
ED-1	QC catch rate	≥95% of issues caught pre-review	PPM manual check comparison
ED-5	Sentiment accuracy	≥85% agreement with PPM assessment	PPM validation sample
CD-1	Retention prediction accuracy	±10% of actual retention	Post-publish comparison

CD-2	Competitor insights used	≥3 insights/week referenced in ideation	CD self-report

11.3 Dashboard Design

Dashboard Section	Widgets	Refresh Rate
Overview	Clients/pod gauge, total agent runs (7d), HITL approval rate, error rate, active users	Real-time
Per-Agent	Runs/day chart, avg latency, tool call distribution, error breakdown, HITL decisions	Hourly
Time Allocation	Current vs. dream bar charts per role (from time audit re-measurements)	Monthly
ROI Tracker	Hours saved/week, capacity value ($), cost vs. value ratio	Weekly
Adoption	DAU, WAU, training completion %, NPS trend	Daily

12. Rollback & Contingency Playbook

12.1 Rollback Scenarios

#	Scenario	Severity	Immediate Action	Recovery
1	**Agent creates wrong task**	MEDIUM	`/rollback [trace_id]` — deletes task, reverts state	Review agent instructions. Add guardrail for the edge case.
2	**Agent assigns wrong editor**	MEDIUM	PC manually reassigns. `/rollback` reverts ClickUp assignment.	Review capacity algorithm weights. Adjust.
3	**Agent sends wrong Slack message**	HIGH	Delete message (Slack API). Notify affected channel.	Tighten HITL gate. Review message templates.
4	**MCP server goes down**	MEDIUM	Auto-failover to direct API Function Tools. Alert in #ai-automation-alerts.	Docker auto-restart. If persistent: manual investigation.
5	**OpenAI API outage**	HIGH	All agents pause. Manual workflows resume. Slack notification to all users.	Monitor OpenAI status page. Resume when restored. No data loss (queued actions retry).
6	**Agent in infinite loop**	MEDIUM	Max iterations guardrail (10 steps) auto-terminates. Alert sent.	Review agent instructions. Add explicit termination conditions.
7	**Data breach concern**	CRITICAL	Immediately disable all agents. Revoke API keys. Notify DFM leadership.	Audit all traces. Identify scope. Remediate. Post-mortem within 24 hours.
8	**Full system failure**	CRITICAL	All agents disabled via feature flag (single toggle). Manual workflows resume immediately.	DFM's existing pipeline works without agents. Zero business continuity risk.

> **Key safety net:** DFM's current pipeline works perfectly without any agents. Agents are additive — they don't replace any existing workflow. If every agent fails simultaneously,

> DFM continues operating exactly as they do today. The worst case is "back to normal," not "broken."

12.2 Feature Flag Architecture

Flag	Default	Controls
`AGENTS_ENABLED`	true	Master kill switch. Disables all agents.
`AGENT_PR4_ENABLED`	true	Per-agent toggle. Disables only PR-4.
`HITL_MODE_PR4`	ALWAYS_CONFIRM	Trust level per agent.
`MCP_CLICKUP_ENABLED`	true	MCP vs. direct API fallback.
`DEMO_MODE`	false	Mock data mode for testing.
`MAX_ACTIONS_PER_RUN`	3	Guardrail: max write operations per agent run.

13. Post-Engagement Sustainability Plan

> **Goal:** After the 6-month engagement, DFM can operate, maintain, and extend the agent infrastructure independently — or with minimal ongoing support.

13.1 Knowledge Transfer Deliverables

#	Deliverable	Format	Delivered
1	**Agent Runbooks**	Notion pages: per-agent setup, configuration, troubleshooting, common issues	Month 5
2	**Architecture Documentation**	GitHub wiki: system diagram, data flow, connector docs, MCP server docs	Month 5
3	**Operations Playbook**	Notion: monitoring, alerting, rollback procedures, escalation paths	Month 6
4	**Training Materials**	Loom library: all training recordings. Notion: quick-reference guides per role.	Ongoing
5	**Code Repository**	GitHub: fully documented, tested, CI/CD configured. README per module.	Continuous
6	**Dashboard Access**	Observability dashboard transferred to DFM-owned infrastructure	Month 6

13.2 Ongoing Support Options

Option	Cost	Includes	Recommended For
Self-Managed	$0	All deliverables above. DFM manages independently.	If DFM hires internal AI/automation engineer
Advisory Retainer	$2K/mo	4 hrs/month: Slack support, architecture guidance, agent tuning, new tool scoping	Most teams
Managed Service	$5K/mo	Full monitoring, agent maintenance, new agent development (up to 2/	If DFM wants to keep scaling

	month), priority support	

13.3 DFM Internal Capability Building

Capability	How We Build It	Timeline
Agent Monitoring	Train DFM stakeholder on dashboard. Document alert response procedures.	Month 4
Agent Tuning	Teach DFM how to modify agent instructions (natural language — no code needed).	Month 5
New Connector Development	Document the connector pattern. If DFM adds a new tool, they can build a connector following the template.	Month 6
Make/Zapier Extension	DFM team already uses Make. Document how agents trigger Make scenarios and vice versa.	Month 3

14. Evaluation Criteria — Final Alignment Across All 6 Batches

> **This section maps every evaluation criterion to specific deliverables across all 6 batches, demonstrating complete coverage of the brief's requirements.**

14.1 Criterion-by-Criterion Mapping

Weight	Criterion	How This Proposal Addresses It	Batch(es)
30%	**Specificity to Your Workflow**	• References all 5 pipeline phases by name (Onboarding → Pre-Pro → Production → Post-Pro → Publish) • Maps every task from the time audit (Feb 4–10, 2026) with exact percentages • Uses DFM tool names and costs ($12,296/mo across 40+ subscriptions) • Addresses all 16 wish-list tools (CD-1–CD-9, ED-1–ED-5, PR-1–PR-4) • References pod structure (Producer, CD, PC, PPM + 10–12 Editors) • Uses dream allocation targets as KPIs — not generic benchmarks • Incorporates AI Tool Survey data (73 respondents, 89% AI adoption) • Names specific DFM team members (Mahmoud, Martina, Hannah, Matias, Garrett, Mariana, Vhinia)	1, 3, 4, 5
25%	**Clarity of Architecture**	• 5-layer architecture (Trigger → Agent → Tool → Integration → HITL) • OpenAI Agents SDK with actual Python code for every agent • MCP availability matrix (ClickUp ✅, Slack ✅, Notion ✅, Frame.io ❌, YouTube ❌) • Hybrid MCP + Function Tool binding code • 47 agent touchpoints across 15 tools • 3 MCP servers, 1	